

Code-Layout, Readability and Reusability

Date	03 july 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Used proper indentation throughout the project.
2	Proper File Structure	Files and folders are logically organized	yes	Organized into frontend, backend, model, dataset and templates folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Clear names like temperature, humidity, crop_prediction.
4	Function / Method Names	Functions are descriptively named	yes	Functions named according to their purpose.
5	Code Comments	Inline and block comments explain logic	yes	Added comments for important functions and ML workflow.
6	Modular Design	Code is split into reusable functions/modules	yes	Prediction, preprocessing and UI handled separately.
7	No Redundant Code	Duplicate or unused code is removed	yes	Removed duplicate code using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	yes	Input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Input Validation	Python, HTML	Form submission	High

2	Prediction Function	Python (Scikit-learn)	Crop prediction	High
3	Model Loader	Pickle	Application startup	High
4	Result Display	HTML, CSS	Prediction page	Medium
5	Data Preprocessing	Pandas	Model prediction	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project folders and files.
Readability	5	Easy to understand with meaningful names.
Reusability	5	Functions reused across the project.
Documentation / Comments	5	Comments added for major modules and logic.
Overall Score	5	Clean, maintainable and modular code.